

Practical 3:- API Integration and Data Rendering in React

Objective:	To consume a REST API in React and handle asynchronous data with loading and error states.
Problem Statement:	• Integrate a public REST API (e.g., GitHub API) into the portfolio application to fetch and display a list of repositories dynamically. • Use the Projects page built in Practical 2 as the integration point. • Implement a loading spinner component shown while the request is in progress. • Implement an error message component shown if the API call fails. • Render at least the repository name and URL for each item returned.
Implementation	Repository List: <pre> import { motion } from 'framer-motion' import { Folder, GitFork, Star, ExternalLink } from 'lucide-react' import { FiGithub } from 'react-icons/fi' const languageColors = { Python: '#3572A5', JavaScript: '#f1e05a', TypeScript: '#3178c6', C: '#555555', 'C++': '#f34b7d', HTML: '#e34c26', CSS: '#563d7c', Jupyter: '#DA5B0B', Dockerfile: '#384d54', } const RepoCard = ({ repo }) => { const { name, html_url: url, description, stargazers_count: stars = 0, forks_count: forks = 0, language, } = repo return (<motion.article whileHover={{ y: -6 }} transition={{ duration: 0.3, ease: [0.22, 1, 0.36, 1] }} className="surface-raised group flex h-full flex-col p-6 transition-shadow duration-300 hover:shadow-hover" > <div className="flex items-start justify-between gap-3"> <div className="flex items-center gap-2.5"> <Folder className="size-4 text-bronze" /> <a </pre>

```

        href={url}
        target="_blank"
        rel="noreferrer"
        className="font-display text-base font-semibold break-all
text-ink transition-colors hover:text-bronze-deep sm:text-lg"
      >
        {name}
      </a>
    </div>
    <span
      className="inline-flex shrink-0 items-center gap-1 rounded-lg
bg-surface px-2.5 py-1 text-xs font-semibold text-ink-soft shadow-soft"
      title="Stars"
    >
      <Star className="size-3.5 fill-bronze text-bronze" />
      {stars.toLocaleString()}
    </span>
  </div>

  <p className="mt-3 flex-1 text-sm leading-relaxed text-ink-soft">
    {description || 'No description provided.'}
  </p>

  <div className="mt-5 flex flex-wrap items-center gap-3 border-t
border-line pt-4">
    {language && (
      <span className="inline-flex items-center gap-1.5 rounded-md
bg-surface px-2.5 py-1 text-[11px] font-medium text-ink-soft shadow-
soft">
        <span
          className="size-2 rounded-full"
          style={{ backgroundColor: languageColors[language] ??
'#a16207' }}
        />
        {language}
      </span>
    )}
    <span
      className="inline-flex items-center gap-1 text-xs font-medium
text-muted"
      title="Forks"
    >
      <GitFork className="size-3.5" />
      {forks.toLocaleString()}
    </span>
    <a
      href={url}
      target="_blank"
      rel="noreferrer"
      className="btn-primary ml-auto inline-flex items-center gap-2
rounded-xl px-4 py-2 text-[13px] font-semibold transition-all duration-
200 hover:-translate-y-0.5"

```

```
        >
        <FiGithub className="size-4" />
        GitHub
        <Externallink className="size-3 opacity-70" />
      </a>
    </div>
  </motion.article>
)
}

const RepoList = ({ repos, hasRepos = true }) => {
  if (repos.length === 0) {
    return (
      <div className="surface-raised px-8 py-14 text-center">
        <p className="text-sm font-medium text-muted">
          {hasRepos
            ? 'No repositories match your search.'
            : 'No public repositories found.'}
        </p>
      </div>
    )
  }

  return (
    <div className="grid gap-6 md:grid-cols-2">
      {repos.map((repo) => (
        <RepoCard key={repo.id} repo={repo} />
      ))}
    </div>
  )
}

export default RepoList
```

	PROJECTS Open-source work A live list of my public repositories, fetched straight from the GitHub API. 24 of 24 repositories Filter repositories by name... EdgeCaseGenerator No description provided. TypeScript 0 GitHub Advanced-Web-Developement-and-Framework No description provided. JavaScript 0 GitHub DataForge A low-code web platform for automated dataset analysis, feature engineering, imbalance handling, and machine learning model recommendation. TypeScript 2 GitHub AgriVision-Analytics No description provided. 0 GitHub repointellect No description provided. TypeScript 0 GitHub super-roast-bot No description provided. 0 GitHub Zaplink_backend No description provided. TypeScript 0 GitHub PDF_QA_Bot No description provided. JavaScript 0 GitHub
Outcome	Students will be able to consume a REST API in React, manage asynchronous state correctly, and provide appropriate UI feedback during all stages of a network request.